

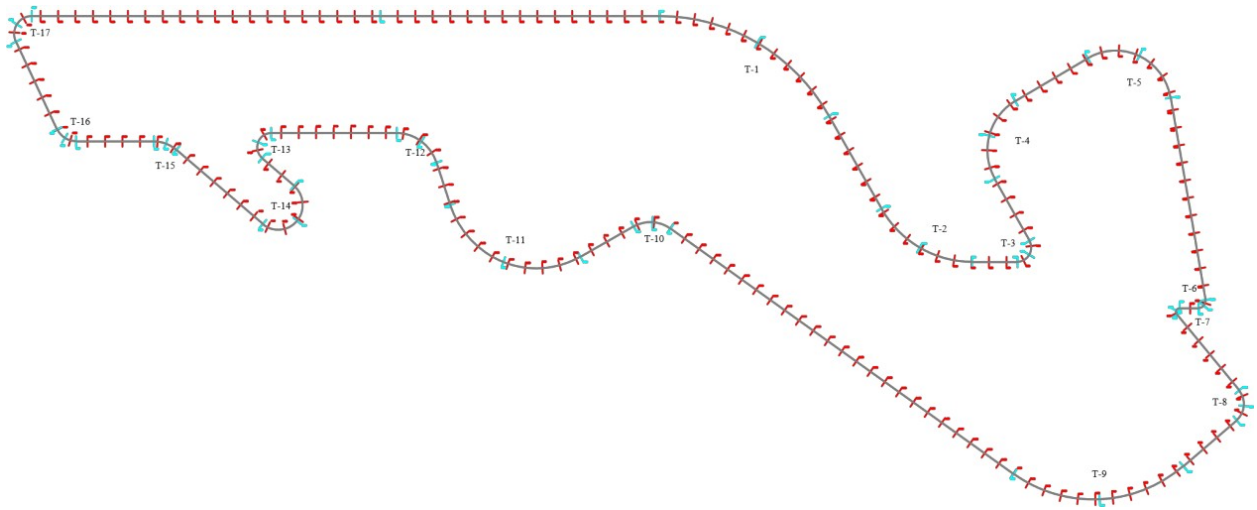
MEF UNIVERSITY  
FACULTY OF ENGINEERING  
DEPARTMENT OF CIVIL ENGINEERING

# Planning and Design of a Competitive Racing Circuit

## Senior Design Project

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*AdanaRing concept layout*

Istanbul 2019

## Executive Summary

This report presents the concept-level planning and engineering design of AdanaRing, a 5,037 m clockwise racing circuit developed as a civil engineering senior design project. The design combines competition objectives with the geometric, drainage, safety, and pavement requirements cited from FIA guidance. Its 17-turn layout is organized into three sectors that alternate between overtaking zones, high lateral-load corners, technical direction changes, and recovery straights.

The design process began with a precedent study and an interview with racing driver and automotive engineer Jason Tahincioğlu. His recommendations established the experiential targets: a total length of 5 to 6 km, a main straight shorter than 1 km, a lap time below approximately 110 seconds, high-load corners, variable-radius combinations, and straights that reward a strong corner exit. These objectives were translated into measurable geometry and checked against the FIA criteria available to the project team.

A speed model was then applied corner by corner using a 325 km/h design top speed, empirical lateral-acceleration data, curve radii, straight lengths, and acceleration and braking estimates. Crossfall was kept within the stated FIA envelope, with a project limit of one percentage point of change per 25 m to protect vehicle floors and reduce abrupt load transfer. Runoff areas were sized for two loss-of-control cases: departure within a corner and departure at the braking point. The second case governed the principal expansions at Turns 3, 10, 11, and other heavy-braking locations.

The pavement concept was checked with the AASHTO structural-number method in WinPAS. The selected inputs produced a required structural number of 2.31; the proposed layer combination produced 3.26. The result supports the feasibility of the preliminary section, but it does not replace site-specific geotechnical investigation, material testing, drainage design, vehicle-dynamics simulation, barrier design, or FIA homologation review.

| Design measure       | Concept value | Design purpose                                               |
|----------------------|---------------|--------------------------------------------------------------|
| Circuit length       | 5,037 m       | Within the project target of 5 to 6 km                       |
| Direction            | Clockwise     | Avoid an unusual sustained neck-load bias for drivers        |
| Turns                | 17            | Mix high-speed, low-speed, and combined corners              |
| Main straight        | 900 m         | Support slipstreaming without exceeding the interview target |
| Track width          | 12 m minimum  | Meet the stated FIA minimum for new permanent circuits       |
| Grid width           | 15 m          | Meet the stated starting-grid requirement                    |
| Design top speed     | 325 km/h      | Set the upper boundary for conceptual speed checks           |
| Required pavement SN | 2.31          | WinPAS output for selected input assumptions                 |
| Provided pavement SN | 3.26          | Preliminary layer combination                                |

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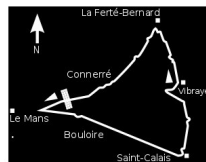
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# 1 Project Definition

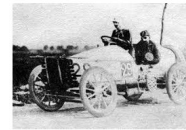
## 1.1 Background

Purpose-built racing circuits emerged because public roads could not safely contain the increasing speed of early motor vehicles. The 1894 Paris to Rouen event emphasized reliability, but the rapid expansion of city-to-city racing exposed drivers, officials, and spectators to uncontrolled roadside hazards. After the 1903 Paris to Madrid race was stopped at Bordeaux following multiple accidents, organized motorsport increasingly separated competition from ordinary traffic.

The 1906 French Grand Prix at Le Mans used a defined triangular course with controlled spectator locations and barriers. This shift established the central design problem that remains relevant: a circuit must create a demanding competition while controlling where vehicles, people, and energy can go when a driver loses control.



Le Mans route 1906



Paris to Madrid 1903

### *Early motorsport and the 1906 Le Mans road circuit*

## 1.2 Objectives

The project had four objectives:

1. Develop a permanent circuit layout that supports close racing, multiple driving styles, and identifiable overtaking zones.
2. Translate the FIA criteria available to the team into track width, straight length, crossfall, grid, and runoff requirements.
3. Estimate corner speeds, braking demands, and runoff distances with a transparent concept-level calculation method.
4. Prepare a preliminary flexible pavement section and verify that its structural number exceeds the calculated requirement.

## 1.3 Scope and Method

The work combines literature review, precedent analysis, expert input, geometric design, simplified vehicle-performance calculations, AutoCAD drafting, and WinPAS pavement analysis. The circuit alignment was assessed at its centerline, divided into three sectors, and evaluated turn by turn. The project uses the track edge rather than an assumed racing line when laying out runoff vectors; this produces a more conservative envelope for the concept stage.

The report is a planning and preliminary design study. It does not establish a construction-ready site design. Final development would require survey control, terrain and earthwork modeling, hydrology and drainage, geotechnical data, materials testing, pit and paddock planning, marshal and medical access, barrier and debris-fence design, spectator sightline and egress studies, noise and environmental assessment, and formal review by the relevant motorsport authority.

## 1.4 Terms Used in the Design

| Term                | Meaning in this report                                                                       |
|---------------------|----------------------------------------------------------------------------------------------|
| Apex                | The inside point of a corner approached by the intended racing line                          |
| Camber or crossfall | The transverse slope of the track surface                                                    |
| On-camber           | Crossfall that assists a vehicle turning toward the inside of the corner                     |
| Off-camber          | Adverse crossfall that tends to move the vehicle toward the outside of the corner            |
| Chicane             | A closely spaced direction change intended to reduce speed and increase technical difficulty |
| Runoff area         | The graded area between the verge and the first line of protection                           |
| Racing line         | The path selected to balance speed, grip, corner entry, apex, and exit                       |
| Lateral G           | Lateral acceleration expressed as a multiple of gravitational acceleration                   |
| ESAL                | Equivalent single axle load used in pavement design                                          |
| Structural number   | AASHTO index representing the combined structural contribution of pavement layers            |

## 2 Design Basis and Precedents

### 2.1 Safety Development

Formula racing developed alongside repeated lessons from high-speed accidents. Vehicle structures, restraint systems, barriers, medical response, and circuit geometry all contribute to safety; no single measure is sufficient. The Mulsanne Straight at Le Mans illustrates this interaction. Group C prototypes exceeded 400 km/h, and fatal accidents occurred during the 1980s. Chicanes were introduced before the 1990 race, while FIA guidance limited the maximum length of a straight to 2 km. The intervention reduced the duration of uninterrupted acceleration and created additional braking and direction-change zones.

The project therefore treats safety as a geometric design responsibility. Corner radius, entry speed, crossfall, braking distance, runoff length, and protection systems must be considered together. The layout must also preserve competitive value, because a circuit that removes every speed or decision point would fail its sporting purpose.

### 2.2 Circuit Precedents

Sixteen circuits were reviewed to identify recurring design responses. The review covered park circuits, street circuits, converted airfields, desert facilities, and permanent purpose-built venues. The most useful lesson was not a single ideal geometry; it was the way each circuit responded to its site, operating environment, and safety history.



Albert Park



Bahrain



Istanbul Park



Monaco

*Examples of park, desert, permanent, and street circuit contexts*

| Circuit           | Length                 | Type               | Design lesson used in the review                                                    |
|-------------------|------------------------|--------------------|-------------------------------------------------------------------------------------|
| Albert Park       | 5.303 km               | Park setting       | Integrates a temporary race venue with an established public landscape              |
| Sepang            | 5.543 km               | Purpose-built      | Combines wide high-speed sections with tighter technical corners                    |
| Imola             | 4.933 km               | Permanent          | Shows how serious accidents can force geometry and protection upgrades              |
| Monaco            | 3.370 km               | Street             | Very limited runoff makes precision, barriers, and qualifying position decisive     |
| Bahrain           | 5.417 km               | Desert             | Addresses sand contamination and high ambient and pavement temperatures             |
| Silverstone       | Not stated             | Converted airfield | Highlights access, drainage, and the long-term adaptation of legacy geometry        |
| Shanghai          | 5.4 km                 | Purpose-built      | Uses symbolic plan geometry and exposes vehicles to acceleration, braking, and wind |
| Monza             | 5.783 km               | Permanent          | Demonstrates the special safety demands of a low-downforce high-speed circuit       |
| Suzuka            | 5.821 km               | Purpose-built      | Uses a figure-eight plan and bridge to create a unique sequence                     |
| Nürburgring       | 20.8 km old 5.1 km new | Permanent          | Shows the emergency-access problem created by long and isolated layouts             |
| Gilles Villeneuve | 4.421 km               | Park or street     | Balances an island park setting with heavy braking and close barriers               |
| Paul Ricard       | 4.26 km                | Permanent          | Evolved through widening, corner revisions, and infrastructure investment           |
| Interlagos        | 4.397 km               | Natural terrain    | Uses topography and a counter-clockwise direction to create a distinct load pattern |
| Hungaroring       | 3.975 km               | Natural bowl       | Uses terrain to improve spectator visibility across much of the circuit             |
| Istanbul Park     | 5.333 km               | Purpose-built      | Uses significant elevation change and widths of approximately 14 to 21 m            |
| Baku              | More than 6 km         | Street             | Combines a narrow heritage section with a long high-speed urban straight            |

## 2.3 FIA Criteria Applied

| Item                  | Cited criterion                                                                  | AdanaRing response                                                                  |
|-----------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Plan geometry         | No fixed plan form; changes may be required for competition or practical reasons | Layout developed freely, then checked against performance and safety objectives     |
| Maximum straight      | 2 km                                                                             | Longest straight limited to 900 m                                                   |
| Circuit measurement   | Measured along the track centerline                                              | Concept length reported as 5,037 m                                                  |
| Track width           | At least 12 m for a new permanent circuit                                        | Minimum width set at 12 m; selected corners widened                                 |
| Width transition      | No faster than 1 m change in 20 m of length                                      | Identified as the governing transition criterion for final CAD revision             |
| Starting grid         | At least 15 m wide                                                               | Grid and start-finish area set to 15 m                                              |
| Start to first corner | Preferably at least 250 m                                                        | Start line positioned to provide more than 250 m before the first qualifying corner |
| Start-finish gradient | Not more than 2 percent                                                          | Longitudinal grades limited to 2 percent or less in the concept                     |
| Straight crossfall    | 1.5 to 3 percent for drainage                                                    | Straight sections held within the stated range                                      |
| Corner banking        | Not more than 10 percent                                                         | Maximum project crossfall is 4 percent                                              |
| Runoff slope          | Up to 25 percent uphill or 3 percent downhill                                    | Concept runoff areas assumed level; final grading remains site dependent            |
| Verge                 | Typically 1 to 5 m with a smooth transition                                      | Continuous verge and runoff transition retained as a design requirement             |

## 2.4 Expert Input

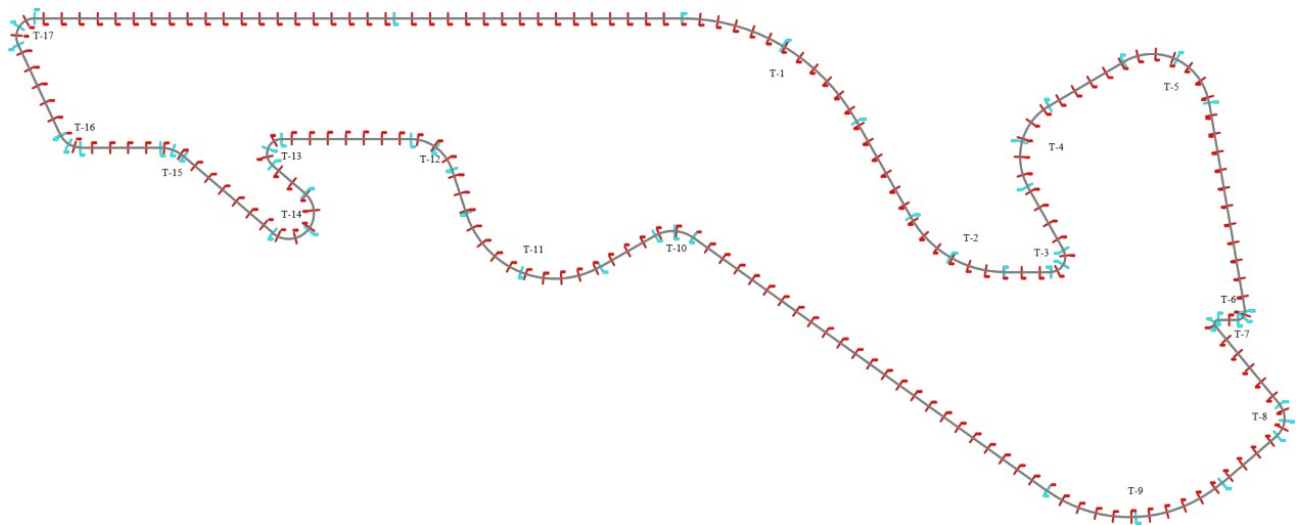
Jason Tahincioğlu was interviewed to connect engineering constraints with the driver's experience. His answers shaped five decisions: keep the circuit between 5 and 6 km; keep the main straight below 1 km; target a lap shorter than about 110 seconds; include high-lateral-load, increasing-radius, and decreasing-radius corners; and place a straight after a combined sequence so that a precise exit produces a measurable reward. The complete edited interview appears in Appendix A.

## 3 Circuit Layout and Competition Logic

### 3.1 Concept Selection

AdanaRing was selected as a clockwise circuit with 17 turns and three sectors. A clockwise direction was preferred because counter-clockwise Formula 1 circuits are less common and can impose unfamiliar sustained loads on drivers. The 5,037 m centerline length sits within the team's 5 to 6 km target, and the 900 m main straight follows the interview recommendation that a primary straight should remain below 1 km.

The layout was not generated by joining isolated favorite corners. Each sequence was assigned a competitive role: create a pass, reward a clean exit, force a compromise between adjacent turns, or provide a short recovery period. This approach created a circuit in which different setups and driving styles can trade time across the lap.



*AdanaRing plan with the 17 numbered turns*

### 3.2 Sector One

Sector One begins on the 900 m main straight. Turn 1 is a 280 m radius, full-throttle right-hander that keeps the slipstream contest active rather than ending the straight with immediate heavy braking. Turn 2 can also be taken at or near full throttle, but its proximity to the 20 m radius Turn 3 creates a decision: carry more speed through Turn 2 and accept a less stable braking phase, or lift earlier and position the car for a controlled entry. The following Turns 4 and 5 have equal radii but different deflection angles. They create sustained lateral load and reward measured throttle application before the first chicane.

The competitive purpose of this sector is to let a following driver remain close through Turn 1, choose a different approach to Turn 2, and attack under braking into Turn 3. A strong Turn 5 exit then determines whether that driver can continue the challenge at Turns 6 and 7.

### 3.3 Sector Two

Turns 6 and 7 form a low-speed chicane. Turn 7 is deliberately off-camber because the available distance is insufficient to reverse the crossfall between the two apexes. This makes car placement more important and prevents the sequence from becoming mechanically predictable. Turns 8 and 9 then open

progressively. Turn 9 is a high-speed, full-throttle corner that rewards smooth steering and early commitment.

A 600 m straight follows Turn 9. It creates a second acceleration zone, allows brake and tire temperatures to stabilize, and gives the driver a brief recovery period before the technical third sector.

### 3.4 Sector Three

Turns 10, 11, and 12 form an esse sequence. The driver must manage repeated direction changes, because an aggressive entry can compromise the following apex and the entire exit. A short straight then leads to the tight Turn 13 overtaking zone. Turn 14 is an off-camber hairpin arranged to help a driver who completes a pass at Turn 13 protect the inside line through the next change of direction.

Turns 15 and 16 create another placement problem, followed by the low-speed final Turn 17. Exit speed from Turn 17 determines performance along the 900 m main straight. A pass at the final corner therefore does not end the contest; the displaced driver gains a slipstream opportunity on the following lap.

### 3.5 Geometry Summary

| Turn | Radius m | Arc length m | Role in the lap                                 |
|------|----------|--------------|-------------------------------------------------|
| T1   | 280      | 293.2        | Full-throttle entry and slipstream continuation |
| T2   | 150      | 157.1        | High-speed setup for the Turn 3 braking zone    |
| T3   | 20       | 41.9         | Primary heavy-braking overtaking corner         |
| T4   | 80       | 125.7        | High-load throttle-management corner            |
| T5   | 80       | 153.5        | Exit quality determines the chicane approach    |
| T6   | 10       | 17.5         | First chicane direction change                  |
| T7   | 5        | 11.3         | Off-camber chicane exit                         |
| T8   | 30       | 47.1         | Low-to-medium-speed transition                  |
| T9   | 200      | 261.8        | Full-throttle high-speed arc                    |
| T10  | 50       | 56.0         | Entry to the esse sequence                      |
| T11  | 101      | 229.0        | Long central arc in the esse sequence           |
| T12  | 60       | 75.2         | Esse exit and setup for Turn 13                 |
| T13  | 20       | 48.9         | Late-braking overtaking corner                  |
| T14  | 35       | 110.0        | Off-camber hairpin for position defense         |
| T15  | 50       | 34.9         | Medium-speed positioning corner                 |
| T16  | 30       | 33.8         | Setup for the final overtaking opportunity      |
| T17  | 25       | 50.4         | Low-speed launch onto the main straight         |

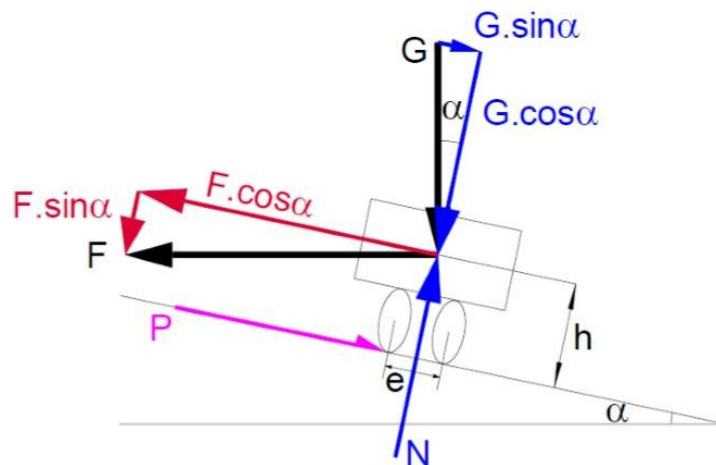
## 4 Speed Model and Corner Assessment

### 4.1 Performance Assumptions

The concept model adopts 325 km/h as the maximum vehicle speed. The value was selected by comparison with Circuit de Barcelona and the speed-trap performance cited for Charles Leclerc. Corner-by-corner estimates require curve radius, arc length, straight length, acceleration, braking deceleration, crossfall, tire-road friction, and aerodynamic load. The team used telemetry-derived lateral-acceleration values as an empirical representation of the additional cornering capability produced by aerodynamic downforce.

| Speed range km/h | Applied lateral acceleration g |
|------------------|--------------------------------|
| 0 to 100         | 2.50                           |
| 100 to 120       | 2.70                           |
| 120 to 150       | 3.00                           |
| 150 to 180       | 3.40                           |
| 180 to 210       | 3.60                           |
| 210 to 240       | 3.90                           |
| 240 to 260       | 4.50                           |
| Above 260        | 4.70                           |

The lateral-acceleration lookup was derived from a 2018 Australian Grand Prix lap attributed to Valtteri Bottas. It provides a consistent concept-level basis for comparing corners, but it is not a full tire or aerodynamic model.



*Forces acting on a vehicle on a banked curve*

## 4.2 Allowable Corner Speed

The project used the following banked-curve expression to estimate the speed at which lateral slip would begin. In the original calculation method, the gravitational term was adjusted using the empirical lateral-acceleration lookup to represent aerodynamic loading.

$$V_{\text{slip}} = \sqrt{gR \left( \frac{\tan \alpha + \mu}{1 - \mu \tan \alpha} \right)}$$

In the equation, R is curve radius,  $\alpha$  is the crossfall angle,  $\mu$  is the tire-road friction coefficient, and g is the acceleration term used by the model. The calculated slip speeds should be read as comparative design outputs. Replacing g directly with a telemetry-derived lateral-G value is a simplification; final design should use a validated tire, aerodynamic, mass, and transient-load model with explicit units and coefficients.

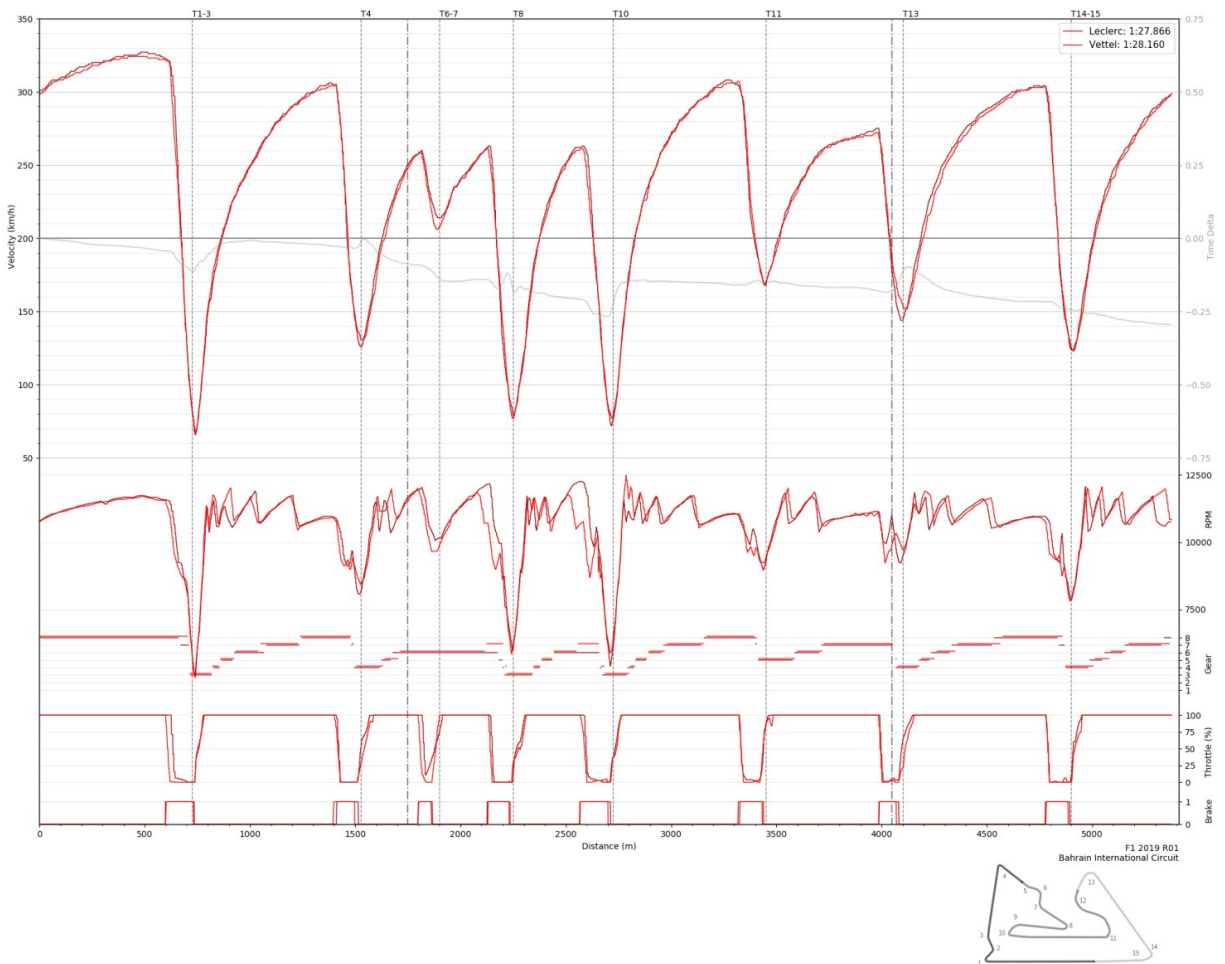
## 4.3 Corner Results

| Turn | Radius m | Calculated slip speed km/h | Design speed km/h | Interpretation                              |
|------|----------|----------------------------|-------------------|---------------------------------------------|
| T1   | 280      | 483                        | 325               | Full throttle within the design top speed   |
| T2   | 150      | 354                        | 325               | Full throttle with early preparation for T3 |
| T3   | 20       | 94                         | 94                | Heavy braking                               |
| T4   | 80       | 253                        | 225 to 253        | Throttle modulation near the limit          |
| T5   | 80       | 253                        | 253               | High-load corner before the chicane         |
| T6   | 10       | 90                         | 90                | Low-speed chicane entry                     |
| T7   | 5        | 40                         | 40                | Very low-speed off-camber exit              |
| T8   | 30       | 126                        | 126               | Progressive acceleration                    |
| T9   | 200      | 409                        | 220 to 280        | Full-throttle high-speed arc                |
| T10  | 50       | 173                        | 170               | Braking from the 600 m straight             |
| T11  | 101      | 264                        | 190               | Mid-esse speed management                   |
| T12  | 60       | 196                        | 196               | Esse exit                                   |
| T13  | 20       | 94                         | 94                | Heavy braking and overtaking                |
| T14  | 35       | 136                        | 136               | Low-speed hairpin                           |
| T15  | 50       | 173                        | 173               | Medium-speed positioning                    |
| T16  | 30       | 120                        | 120               | Setup for the final corner                  |
| T17  | 25       | 109                        | 109               | Exit speed governs the main straight        |

## 4.4 Acceleration and Braking

Acceleration was represented in 25 km/h increments. The source data show strong low-speed acceleration that reduces as aerodynamic drag increases. Braking scenarios were evaluated from the predicted speed at each braking point to the target corner speed. For runoff sizing, telemetry from the 2019 Bahrain Grand Prix qualifying session produced an average braking deceleration of 22.775 m/s<sup>2</sup>, approximately 2.32 g.

These values are suitable for a comparative concept study. They do not account for tire condition, fuel load, weather, brake temperature, aerodynamic damage, driver reaction, vehicle rotation, locked wheels, or the lower deceleration available after a mechanical failure. Those cases should be included in a formal safety simulation.



*Qualifying telemetry used as a reference for acceleration and braking estimates*

## 5 Crossfall and Superelevation Design

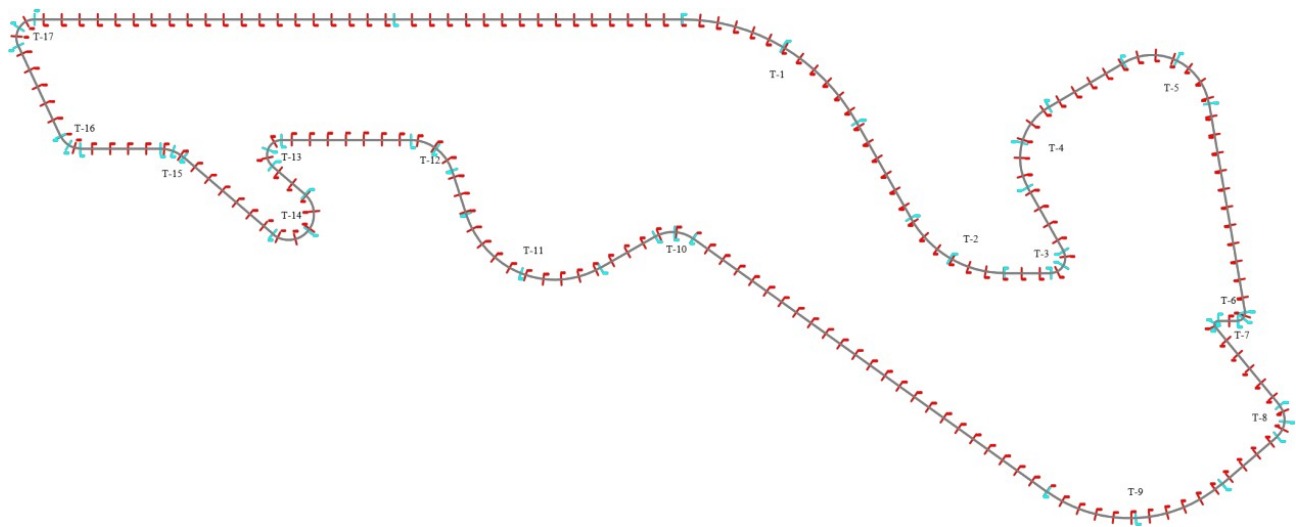
### 5.1 Design Envelope

The crossfall design had to satisfy drainage while avoiding abrupt changes that could strike the vehicle floor or create sudden load transfer. The cited FIA criteria require 1.5 to 3 percent crossfall on straights, limit corner banking to 10 percent, and restrict an off-camber corner with an entry speed above 125 km/h. AdanaRing uses crossfalls between minus 4 and plus 4 percent, well below the maximum banking limit.

The project added a self-imposed transition limit of one percentage point per 25 m. This limit is not presented as an FIA rule; it is a design control selected to make crossfall changes gradual and predictable.

### 5.2 Crossfall Strategy

On-camber corners were used where the layout should support a stable, high-speed line. Off-camber geometry was reserved for low-speed locations where additional technical difficulty would not conflict with the stated 125 km/h entry limit. The principal examples are Turn 7 at the chicane exit and Turn 14 after the Turn 13 overtaking zone.



*AdanaRing crossfall control points along the alignment*

### 5.3 Segment Schedule

| Segment       | Start percent | End percent | Design intent                                       |
|---------------|---------------|-------------|-----------------------------------------------------|
| Main straight | +3            | +3          | Constant drainage crossfall                         |
| T1            | +3            | 0           | Reduces support before the direction change into T2 |
| T1 to T2      | 0             | -3          | Prepares the T2 and T3 sequence                     |
| T2            | -3            | -3          | Constant through the high-speed corner              |
| T2 to T3      | -3            | -4          | Additional on-camber support for the braking corner |

| Segment    | Start percent | End percent | Design intent                                      |
|------------|---------------|-------------|----------------------------------------------------|
| T3         | -4            | -4          | Constant through the overtaking zone               |
| T3 to T4   | -4            | 0           | Removes adverse crossfall before the fast T4 entry |
| T4         | 0             | +1          | Gradual support through the corner                 |
| T4 to T5   | +1            | +4          | Prepares the longer T5 arc                         |
| T5 to T7   | +4            | +4          | Maintains one plane through the chicane            |
| T7         | +4            | +4          | Off-camber left-hander at low speed                |
| T7 to T8   | +4            | +2          | Reduces crossfall before T8                        |
| T8         | +2            | +2          | Constant through the 90-degree corner              |
| T8 to T9   | +2            | +3          | Supports the high-speed T9 arc                     |
| T9         | +3            | +3          | Constant high-speed crossfall                      |
| T9 to T10  | +3            | -2          | Long transition along the 600 m straight           |
| T10 to T12 | -2            | -2          | Multiple controlled reversals through the esses    |
| T12 to T13 | -2            | -2          | Constant on the approach                           |
| T13 to T14 | -2            | -4          | Creates the off-camber T14 hairpin                 |
| T14 to T15 | -4            | -2          | Reduces adverse crossfall                          |
| T15        | -2            | -2          | On-camber left-hander                              |
| T15 to T16 | -2            | +2          | Reverses crossfall for the right-hander            |
| T16        | +2            | +2          | Constant through the corner                        |
| T16 to T17 | +2            | +3          | Prepares the final corner and main straight        |
| T17        | +3            | +3          | Constant through the final corner                  |

The detailed design notes calculate transition rates between 0.18 and 0.90 percentage points per 25 m for the listed straight sections. All remain within the team's one-percentage-point project limit.

## 6 Runoff Area Design

### 6.1 Safety Scenarios

Runoff geometry was developed after the speed and crossfall assessment. The concept assumes smooth transitions from the track to the verge and runoff area, with no adverse grade in the preliminary plan. Two loss-of-control scenarios were considered.

1. Corner departure. The vehicle leaves the track while traveling at the maximum allowable speed assigned to the curve. Vectors are projected from the track edge through the corner, and their endpoints define the base runoff envelope.
2. Braking-point departure. The vehicle loses control before reaching the target corner speed. The envelope is expanded using the higher predicted speed at the braking point and the corresponding stopping distance.

The second scenario governs most heavy-braking zones because the vehicle carries substantially more kinetic energy before deceleration begins. A 10 percent speed margin was applied in the original stopping-time calculations.

### 6.2 Stopping Distance Basis

For constant deceleration, the idealized stopping distance is expressed as follows.

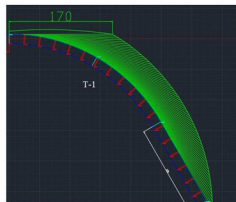
$$d_{\text{stop}} = \frac{v^2}{2a}$$

The project used an average braking deceleration of 22.775 m/s<sup>2</sup> obtained from the cited Bahrain telemetry. The AutoCAD envelopes were drawn from the track edge rather than from a single racing line. This choice intentionally reserves the pavement width between the normal line and the edge as additional tolerance rather than counting it as runoff.

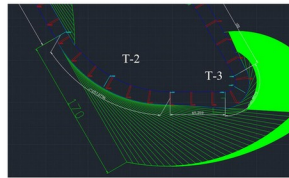
### 6.3 Turn by Turn Results

| Turn     | Allowable speed km/h | Braking point speed km/h | Stopping basis m | Design response                                         |
|----------|----------------------|--------------------------|------------------|---------------------------------------------------------|
| T1       | 483                  | 325                      | 170              | Base envelope sufficient at the design top speed        |
| T2       | 354                  | 325                      | 170              | Base envelope sufficient; coordinated with T3           |
| T3       | 94                   | 325                      | 25 and 170       | Expanded for braking-point departure and joined to T2   |
| T4       | 253                  | 300                      | 105              | Shared envelope with T5                                 |
| T5       | 253                  | 245                      | 105 to 110       | Base envelope retained                                  |
| T6 to T7 | 90 and 40            | 245                      | 110 to 140       | Chicane envelope extended toward the following straight |
| T8       | 126                  | 189                      | 30 and 75        | Expanded for approach speed                             |

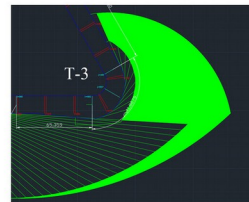
| Turn | Allowable speed km/h | Braking point speed km/h | Stopping basis m | Design response                                    |
|------|----------------------|--------------------------|------------------|----------------------------------------------------|
| T9   | 409                  | 280                      | 130              | Base envelope sufficient for the full-throttle arc |
| T10  | 173                  | 325                      | 50 and 170       | Major expansion after the 600 m straight           |
| T11  | 264                  | 234                      | 50 and 90        | Expanded for the esse approach                     |
| T12  | 196                  | 250                      | 65 and 100       | Expanded for braking-point departure               |
| T13  | 94                   | 255.8                    | 15 and 110       | Major expansion at the overtaking corner           |
| T14  | 125                  | 165.3                    | 25 and 50        | Expanded around the hairpin exit direction         |
| T15  | 173                  | 229.4                    | 50 and 80        | Expanded for approach speed                        |
| T16  | 120                  | 221                      | 25 and 75        | Expanded before the final sequence                 |
| T17  | 109                  | 230                      | 20 and 80        | Expanded for the final-corner braking point        |



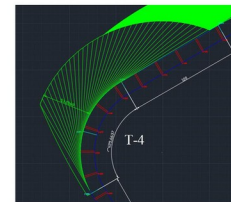
T01



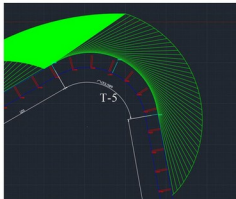
T02



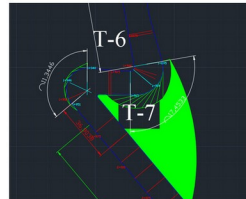
T03



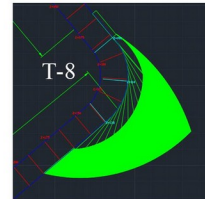
T04



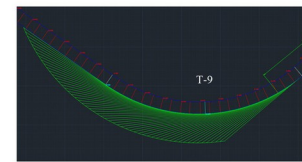
T05



T06\_07

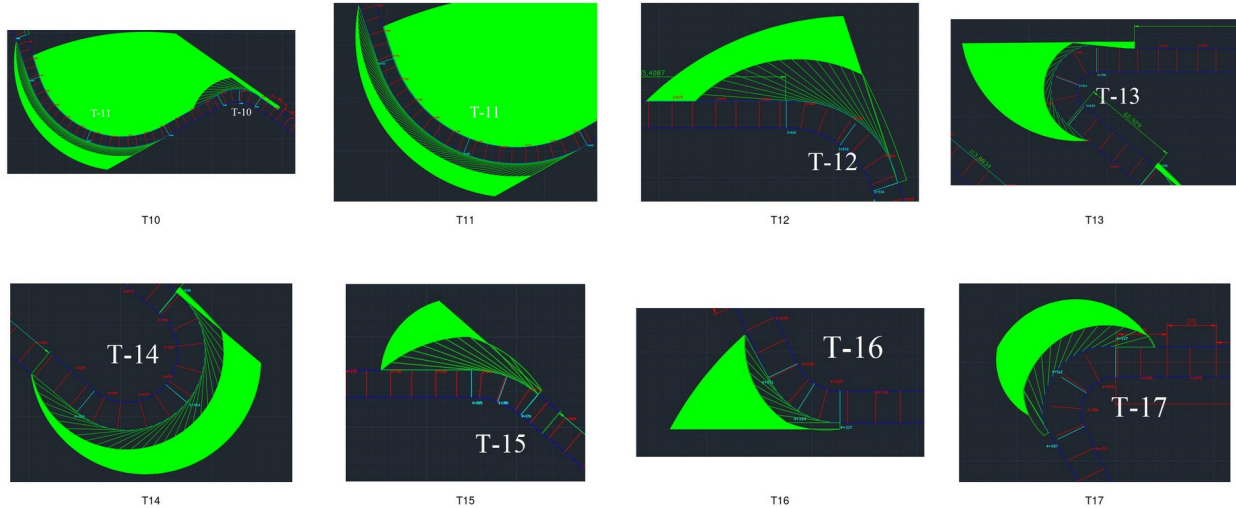


T08



T09

*AutoCAD runoff envelopes for Turns 1 to 9*



*AutoCAD runoff envelopes for Turns 10 to 17*

## 6.4 Width Transitions and Design Reconciliation

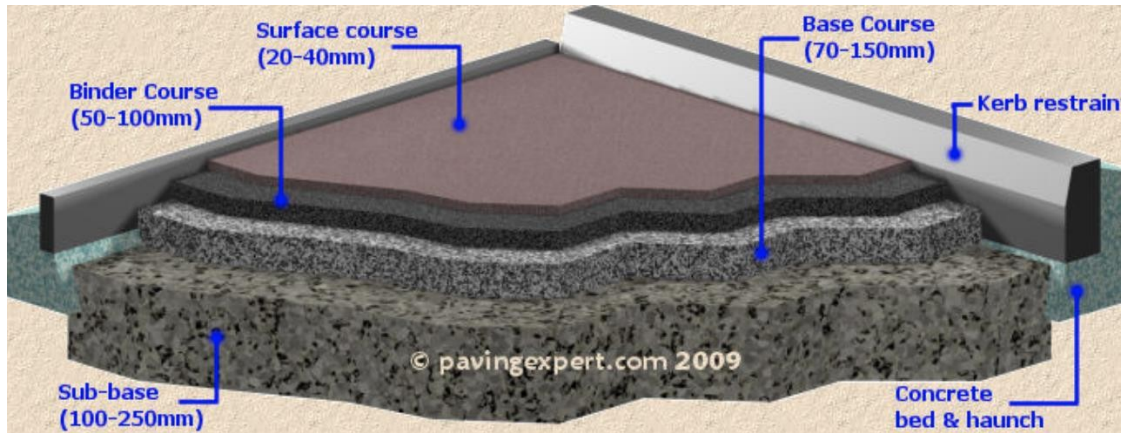
The cited FIA criterion limits a width change to 1 m over at least 20 m of longitudinal distance. The original concept notes also refer to a 1:10 transition observed at Istanbul Park. These two values are not equivalent. The 1:20 criterion should govern the next CAD revision unless a current authority-approved exception applies. The widening dimension at Turn 3 should also be confirmed directly from the CAD model before issue, because the original narrative and transition length are not internally consistent.

This reconciliation does not change the location of the governing runoff zones. It changes how gradually the track and paved shoulder must reach their final width.

## 7 Pavement Design

### 7.1 Pavement Concept

A flexible pavement section was selected for the concept. The surface course provides the riding surface, texture, drainage path, and protection of lower layers. The binder and base courses distribute load and provide structural support, while the granular layers support drainage and separate the asphalt system from the subgrade.



*Typical flexible pavement structure used as the design basis*

### 7.2 AASHTO Structural Number

The AASHTO structural number combines layer coefficients, thicknesses, and drainage coefficients.

$$SN = a_1D_1 + a_2D_2m_2 + a_3D_3m_3$$

Here,  $a$  represents the structural coefficient of each layer,  $D$  is layer thickness in inches, and  $m$  is the drainage coefficient for an unbound layer. WinPAS was used to calculate the required structural number and to compare it with the selected layer combination.

### 7.3 Design Inputs and Result

| Input                      | Value       | Basis stated in the project                      |
|----------------------------|-------------|--------------------------------------------------|
| Total flexible ESALs       | 100,000     | Concept-level traffic assumption                 |
| Reliability                | 95 percent  | High reliability selected for the racing surface |
| Overall standard deviation | 0.45        | Value used for new asphalt pavement              |
| Subgrade resilient modulus | 8,558.1 psi | Derived from selected average CBR and R values   |
| Initial serviceability     | 5.0         | New surface condition                            |
| Terminal serviceability    | 2.0         | Minimum acceptable serviceability in the model   |

| Input                      | Value | Basis stated in the project |
|----------------------------|-------|-----------------------------|
| Required structural number | 2.31  | WinPAS output               |

| Layer                   | Coefficient | Drainage factor | Thickness in | SN contribution |
|-------------------------|-------------|-----------------|--------------|-----------------|
| Asphalt cement concrete | 0.30        | 1.00            | 5.0          | 1.50            |
| Crushed stone base      | 0.12        | 1.00            | 5.5          | 0.66            |
| Graded stone base       | 0.20        | 1.00            | 5.5          | 1.10            |
| Granular subbase        | 0.00        | 1.00            | 5.0          | 0.00            |
| Total                   |             |                 | 21.0         | 3.26            |

The proposed structural number of 3.26 exceeds the required value of 2.31 by 0.95. On the assumptions entered in WinPAS, the preliminary pavement section therefore passes the structural-number check.

|                         |                     |                      | inches           |                     | inches             |
|-------------------------|---------------------|----------------------|------------------|---------------------|--------------------|
| Layer Material          | Layer Coefficien... | Drainage Coeffici... | Layer Thickne... | Layer Struct No,... | Additional Thic... |
|                         |                     |                      |                  | 0                   | -3.17              |
|                         |                     |                      |                  | 0                   | -3.17              |
| Asphalt Cement Concrete | 0.3                 | 1                    | 5                | 1.5                 | -3.17              |
| Crushed Stone Base      | 0.12                | 1                    | 5.5              | 0.66                | -7.92              |
| Graded Stone Base       | 0.2                 | 1                    | 5.5              | 1.1                 | -4.75              |
| Granular Subbase        | 0                   |                      | 5                | 0                   | -7.92              |
| Sum of SN               |                     |                      |                  | 3.26                | OK                 |
| SN                      |                     |                      |                  | 2.31                |                    |

*WinPAS layer contributions and structural-number comparison*

## 7.4 Pavement Limitations

The pavement calculation is preliminary. A racing surface is controlled by more than axle-load capacity: smoothness, macrotexture, microtexture, wet friction, thermal cycling, shear under braking, fuel and oil exposure, joint detailing, drainage, and repair tolerances all affect performance. The zero structural contribution assigned to the granular subbase should be reviewed, and the selected layer coefficients should be verified against tested local materials. The final design should also distinguish racing loads from construction, maintenance, emergency, and service vehicles.

## 8 Conclusions and Development Path

### 8.1 Design Outcome

The project produced a coherent concept for a competitive racing circuit. AdanaRing meets the principal geometry targets established by the team: a 5,037 m clockwise layout, a 900 m main straight, 17 turns, a 12 m minimum track width, a 15 m starting grid, and crossfalls within the cited FIA envelope. Its three sectors create distinct performance problems rather than repeating the same corner type.

The most important design decision was to connect the sporting objective to the engineering checks. The Turn 2 to Turn 3 sequence creates a braking decision, the Turn 6 to Turn 7 chicane uses low-speed adverse crossfall to add difficulty, the Turn 10 to Turn 12 esses reward consistency, and the Turn 13 to Turn 14 sequence links an overtaking opportunity to a defensive line. Runoff sizing then tests the consequences of those decisions at the predicted speeds.

### 8.2 Engineering Responsibility

Driver safety, spectator safety, marshal access, and emergency response must remain design requirements throughout development. A fast and entertaining layout cannot compensate for an insufficient safety envelope. Equally, a calculation that appears conservative is not a substitute for a documented model, validated inputs, and independent review. The inconsistencies identified in the width-transition note and the simplified vehicle model should be resolved before the design is used beyond an academic concept.

### 8.3 Learning and Teamwork

The project required the team to work across civil, automotive, and motorsport disciplines. Road geometry, drainage, pavement design, vehicle dynamics, CAD drafting, telemetry interpretation, and safety planning are interdependent. The work reinforced a practical lesson: engineering judgment improves when assumptions are written clearly enough for another person to challenge them.

The design phase was the most demanding part of the project because no single source provided a complete recipe for a racing circuit. The team responded by combining formal criteria with precedents, expert input, and conservative geometric choices. That process also revealed where future students and designers would benefit from a more integrated design guideline.

### 8.4 Recommended Next Phase

1. Select and survey a site; integrate terrain, earthworks, drainage, and environmental constraints.
2. Validate the layout with a transient vehicle model covering tires, aerodynamics, braking failures, weather, and multiple vehicle classes.
3. Reconcile every track-width transition with the governing 1:20 criterion and verify the CAD dimensions.
4. Design barriers, debris fencing, marshal posts, rescue routes, medical access, and spectator separation to current requirements.
5. Develop pit lane, paddock, race control, utilities, transport, parking, egress, and event operations as one site plan.
6. Use geotechnical investigation and laboratory testing to finalize pavement materials and layer thicknesses.

## References

1. Fédération Internationale de l'Automobile. Procedures for the Recognition of Motor Racing Circuits. Clauses cited in the original project report for plan geometry, width, grade, crossfall, grid, and runoff requirements.
2. Fuller, Michael J. 2010. Mulsanne's Corner Maximum Speeds at Le Mans 1961 to 1989.
3. Speedhunters staff. 2008. Material cited in the original report regarding Mulsanne Straight and the addition of chicanes.
4. Heimbs et al. 2009. Study cited in the original report regarding the relationship between vehicle and circuit safety.
5. Kuhn, Wolfgang. 2016. Interactive driving-simulation methodology for training-circuit design, as cited in the original report.
6. Formula 1 official video and telemetry material. 2018 Australian Grand Prix lateral-acceleration values used by the project team.
7. 2019 Bahrain Grand Prix qualifying telemetry for Charles Leclerc and Sebastian Vettel. Braking values used by the project team.
8. American Association of State Highway and Transportation Officials. Flexible pavement structural-number method.
9. WinPAS pavement design and analysis software. Structural-number calculation used in the project.
10. Tahincioğlu, Jason. Interview conducted for the project. Edited transcript reproduced in Appendix A.

## Appendix A Expert Interview

Jason Tahincioğlu is a Turkish racing driver and automotive engineer. He began karting at the age of six, competed in GP2 during the 2006 and 2007 seasons, and tested a Toyota Jordan Formula 1 car. The following transcript has been edited for grammar while preserving the substance of his answers.

### A.1 Interview Transcript

#### **Which corners or corner combinations do you enjoy most**

I particularly like the Bomb Hole at Snetterton. A GP2 car could take it at full throttle, which produced a high lateral load. The following corner could also be taken flat, and that made the sector one of my favorites.

#### **What is the ideal length of a main straight**

It should be shorter than 1 km. Longer straights become boring.

#### **What is the ideal total circuit length**

Between 5 and 6 km. In my opinion, a lap should remain below about 110 seconds. Drivers have limited track time during practice; when lap times become too long, they complete fewer laps and have less opportunity to build the experience and confidence needed to show their ability.

#### **What makes a circuit technical**

A technical circuit should force drivers to negotiate between adjacent corners. There should be combinations where gaining time in one turn costs time in the next. The circuit should support different driving styles and allow a driver to improve with every lap. Blind corners such as Turn 3 at Istanbul Park or Raidillon at Spa-Francorchamps also test a driver's ability. A technical circuit should allow changes in strategy, driving style, and racing line while remaining extremely demanding.

#### **Is there anything else you would add**

A circuit should include increasing-radius and decreasing-radius corners. It should also include straights that let drivers recover. A combined sequence should lead onto a straight, because the straight becomes the reward for the driver who completes the sequence more effectively.

## Appendix B Core Design Tables

### B.1 Straight Lengths

| Interval    | Length m |
|-------------|----------|
| Start to T1 | 400      |
| T1 to T2    | 150      |
| T2 to T3    | 65       |
| T3 to T4    | 100      |
| T4 to T5    | 120      |
| T5 to T6    | 288      |
| T6 to T7    | 27       |
| T7 to T8    | 139      |
| T8 to T9    | 100      |
| T9 to T10   | 600      |
| T10 to T11  | 87       |
| T11 to T12  | 62       |
| T12 to T13  | 180      |
| T13 to T14  | 60       |
| T14 to T15  | 161      |
| T15 to T16  | 111      |
| T16 to T17  | 136      |
| T17 to T1   | 900      |

### B.2 Acceleration Data

| Speed interval km/h | Time increment s | Average acceleration m/s <sup>2</sup> |
|---------------------|------------------|---------------------------------------|
| 0 to 25             | 0.53             | 13.00                                 |
| 25 to 50            | 0.63             | 10.95                                 |
| 50 to 100           | 0.50             | 13.78                                 |
| 100 to 150          | 0.47             | 14.74                                 |
| 150 to 200          | 0.94             | 7.40                                  |
| 200 to 250          | 1.07             | 6.51                                  |
| 250 to 325          | 1.80             | 3.85                                  |